

Water Quality Monitoring System

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Abstract

This paper presents the design and implementation of an automated water quality monitoring system using the ESP32 microcontroller. The system is capable of measuring temperature and Total Dissolved Solids (TDS) in water, displaying data on an SSD1306 OLED display, and connecting via WiFi to transmit data to the ThingSpeak platform for remote monitoring and analysis. The system is designed with low cost, easy deployment, and scalability for IoT applications in water quality monitoring.

Keywords: ESP32, water quality monitoring, TDS, temperature, IoT, ThingSpeak, SSD1306

1 Introduction

Water quality is a critical factor that directly affects human health and the environment. Continuous monitoring of water quality parameters such as temperature and Total Dissolved Solids (TDS) is essential to ensure water meets usage standards. TDS is an important indicator that reflects the total amount of inorganic and organic substances dissolved in water, including ions such as sodium, potassium, calcium, magnesium, and other compounds.

With the development of Internet of Things (IoT), building automated monitoring systems and transmitting data to the cloud has become more feasible and efficient. ESP32 is a powerful microcontroller with integrated WiFi and Bluetooth, suitable for IoT applications. ThingSpeak is an open-source IoT platform that allows collection, storage, and analysis of data from sensor devices.

This paper presents the design and implementation of a water quality monitoring system using ESP32, with the following main features:

- Water temperature measurement using temperature sensor
- TDS measurement using TDS sensor
- Data display on SSD1306 OLED screen
- WiFi connection and data transmission to ThingSpeak

2 System Overview

2.1 System Architecture

The system is designed with a simple and efficient architecture, consisting of the following main components:

- **ESP32:** Main microcontroller, processes data and manages WiFi connection
- **DS18B20 Temperature Sensor:** Measures water temperature with high accuracy
- **TDS Sensor:** Measures total dissolved solids in water
- **SSD1306 OLED Display:** Displays measured data in real-time
- **Integrated WiFi Module:** Connects to Internet and sends data to ThingSpeak

2.2 Operation Flow

The system operates according to the following steps:

1. Initialize ESP32 and initialize components (sensors, OLED display)
2. Connect to WiFi with configured credentials
3. Read data from temperature sensor and TDS sensor
4. Display data on OLED screen
5. Send data to ThingSpeak via HTTP protocol
6. Repeat the process after a periodic interval (e.g., 15 seconds)

3 Hardware Design

3.1 ESP32 Development Board

ESP32 is a 32-bit microcontroller with the following main characteristics:

- CPU: Dual-core Xtensa LX6, 240 MHz clock speed
- Memory: 520 KB SRAM, supports external flash
- Connectivity: WiFi 802.11 b/g/n, Bluetooth 4.2
- GPIO: 34 programmable GPIO pins
- ADC: 12-bit ADC with 18 channels
- Operating voltage: 3.3V

ESP32 was chosen due to its integrated WiFi, strong processing capability, and reasonable cost.

3.2 DS18B20 Temperature Sensor

DS18B20 is a digital temperature sensor with the following characteristics:

- Measurement range: -55°C to $+125^{\circ}\text{C}$
- Accuracy: $\pm 0.5^{\circ}\text{C}$ (from -10°C to $+85^{\circ}\text{C}$)
- Resolution: 9 to 12 bits (configurable)
- Communication: OneWire protocol
- Voltage: 3.0V to 5.5V

The sensor is connected to ESP32 via GPIO pin and uses OneWire and DallasTemperature libraries to read data.

3.3 TDS Sensor

The TDS sensor measures the electrical conductivity of water to calculate total dissolved solids. Operating principle:

- Measures electrical conductivity of water solution
- Converts analog voltage value to digital value
- Calculates TDS based on formula: $\text{TDS (ppm)} = k \times \text{EC (S/cm)}$
- Coefficient k is typically 0.5-0.7 depending on the type of dissolved solids

The TDS sensor is connected to ESP32 via ADC pin to read analog voltage value.

3.4 SSD1306 OLED Display

SSD1306 is a 128x64 pixel OLED display with the following characteristics:

- Size: 128x64 pixels
- Communication: I2C or SPI
- Voltage: 3.3V or 5V
- Low power consumption

The display is connected to ESP32 via I2C communication (SDA, SCL) to display measured data.

4 Software Design

4.1 Development Environment

The system is programmed in C++ using Arduino IDE with the following libraries:

- `WiFi.h`: Manages WiFi connection
- `HTTPClient.h`: Sends HTTP data to ThingSpeak
- `OneWire.h`: Communicates with DS18B20 sensor
- `DallasTemperature.h`: Reads temperature from DS18B20
- `Wire.h`: I2C communication with OLED display
- `Adafruit_SSD1306.h`: Controls OLED display
- `Adafruit_GFX.h`: Graphics library for OLED

4.2 Program Structure

The main program is organized into functional functions:

Listing 1: Main program structure

```
1 void setup() {
2     // Initialize Serial Monitor
3     Serial.begin(115200);
4
5     // Initialize OLED display
6     initOLED();
7
8     // Initialize temperature sensor
9     initTemperatureSensor();
10
11    // Connect WiFi
12    connectWiFi();
13
14    // Display startup message
15    displayStartupMessage();
16 }
17
18 void loop() {
19     // Read temperature
20     float temperature = readTemperature();
21
22     // Read TDS
23     float tds = readTDS();
24
25     // Display on OLED
26     displayData(temperature, tds);
27
28     // Send to ThingSpeak
```

```

29     sendToThingSpeak(temperature, tds);
30
31     // Wait 15 seconds
32     delay(15000);
33 }

```

4.3 WiFi Connection

The WiFi connection function is implemented as follows:

Listing 2: WiFi connection

```

1 void connectWiFi() {
2     WiFi.mode(WIFI_STA);
3     WiFi.begin(ssid, password);
4
5     Serial.print("Connecting to WiFi");
6     while (WiFi.status() != WL_CONNECTED) {
7         delay(500);
8         Serial.print(".");
9     }
10
11     Serial.println();
12     Serial.print("Connected! IP address: ");
13     Serial.println(WiFi.localIP());
14 }

```

4.4 Sensor Data Reading

4.4.1 Temperature Reading

Listing 3: Reading temperature from DS18B20

```

1 float readTemperature() {
2     sensors.requestTemperatures();
3     float temp = sensors.getTempCByIndex(0);
4
5     if (temp == DEVICE_DISCONNECTED_C) {
6         Serial.println("Error: Could not read temperature");
7         return -999.0;
8     }
9
10    return temp;
11 }

```

4.4.2 TDS Reading

Listing 4: Reading TDS value

```

1 float readTDS() {
2     int analogValue = analogRead(TDS_PIN);

```

```

3     float voltage = analogValue * (3.3 / 4095.0);
4
5     // Calculate TDS (formula can be adjusted)
6     float tdsValue = (133.42 * voltage * voltage * voltage
7                       - 255.86 * voltage * voltage
8                       + 857.39 * voltage) * 0.5;
9
10    return tdsValue;
11 }

```

4.5 Sending Data to ThingSpeak

ThingSpeak is an IoT platform that allows data collection and analysis. To send data, the following are required:

- API Key from ThingSpeak channel
- Channel ID (if using private channel)

Listing 5: Sending data to ThingSpeak

```

1 void sendToThingSpeak(float temperature, float tds) {
2     if (WiFi.status() == WL_CONNECTED) {
3         HTTPClient http;
4
5         String url = "http://api.thingspeak.com/update?api_key="
6                     + String(THINGSPEAK_API_KEY)
7                     + "&field1=" + String(temperature)
8                     + "&field2=" + String(tds);
9
10        http.begin(url);
11        int httpResponseCode = http.GET();
12
13        if (httpResponseCode > 0) {
14            Serial.print("HTTP Response code: ");
15            Serial.println(httpResponseCode);
16        } else {
17            Serial.print("Error code: ");
18            Serial.println(httpResponseCode);
19        }
20
21        http.end();
22    }
23 }

```

4.6 OLED Display

Listing 6: Displaying data on OLED

```

1 void displayData(float temperature, float tds) {
2     display.clearDisplay();

```

```

3   display.setTextSize(1);
4   display.setTextColor(SSD1306_WHITE);
5
6   // Display title
7   display.setCursor(0, 0);
8   display.println("Water Quality Monitor");
9   display.drawLine(0, 10, 128, 10, SSD1306_WHITE);
10
11  // Display temperature
12  display.setCursor(0, 15);
13  display.print("Temperature: ");
14  display.print(temperature);
15  display.println(" C");
16
17  // Display TDS
18  display.setCursor(0, 30);
19  display.print("TDS: ");
20  display.print(tds);
21  display.println(" ppm");
22
23  // Display WiFi status
24  display.setCursor(0, 45);
25  if (WiFi.status() == WL_CONNECTED) {
26      display.println("WiFi: Connected");
27  } else {
28      display.println("WiFi: Disconnected");
29  }
30
31  // Display time
32  display.setCursor(0, 55);
33  display.print("Time: ");
34  display.print(millis() / 1000);
35  display.println(" s");
36
37  display.display();
38 }

```

5 Experimental Results

5.1 Test Configuration

The system was tested under the following conditions:

- Power supply: 5V via USB port or adapter
- WiFi range: Within 10m from router
- Measurement cycle: Every 15 seconds
- Water samples: Tap water, distilled water, mineral water

5.2 Measurement Results

Table 1 presents the measurement results from the system with different water samples.

Table 1: Water quality measurement results

Water Sample	Temperature (°C)	TDS (ppm)
Distilled water	25.2	5
Tap water	24.8	145
Mineral water	25.0	320

5.3 Performance Evaluation

The system achieved the following specifications:

- Temperature measurement accuracy: $\pm 0.5^{\circ}\text{C}$
- TDS measurement accuracy: ± 10 ppm (in range 0-1000 ppm)
- Response time: ≤ 2 seconds
- Success rate of sending data to ThingSpeak: $\geq 95\%$
- Power consumption: 150mA at 5V (during operation)

5.4 ThingSpeak Display

Data is sent to ThingSpeak and displayed as real-time charts. Users can:

- View data history
- Analyze water quality change trends
- Set alerts when values exceed thresholds
- Export data for deeper analysis

6 Discussion

6.1 System Advantages

- **Low cost:** Uses common components with reasonable prices
- **Easy deployment:** Simple configuration, no complex infrastructure required
- **Remote monitoring:** Data is sent to cloud, accessible from anywhere
- **Scalable:** Can add other sensors such as pH, turbidity, dissolved oxygen
- **Visual display:** OLED screen allows on-site data viewing

6.2 Limitations and Future Development

- **WiFi dependency:** System requires stable WiFi connection to send data
- **Power supply:** Requires stable power source, can integrate battery for independent operation
- **TDS accuracy:** Requires periodic calibration to ensure accuracy
- **Sensor protection:** Requires waterproof enclosure for sensors when used outdoors

Future development directions:

- Integrate additional sensors: pH, turbidity, dissolved oxygen (DO)
- Use battery and solar power for independent operation
- Develop mobile application for data monitoring
- Integrate machine learning for water quality prediction
- Use LoRaWAN or GSM for data transmission in areas without WiFi

7 Conclusion

This paper has presented the successful design and implementation of a water quality monitoring system using ESP32 with temperature and TDS measurement features, OLED display, and ThingSpeak data transmission. The system has low cost, easy deployment, and is suitable for IoT applications in water quality monitoring.

Test results show that the system operates stably with acceptable accuracy. Data is successfully sent to ThingSpeak and can be monitored remotely. The system has potential for expansion to integrate additional sensors and improve features.

Figure 1 shows the PCB layout design of the system, illustrating the routing of traces and component placement. The layout demonstrates an organized design with proper connections between different components including the ESP32 module, sensor interfaces, and display connections.

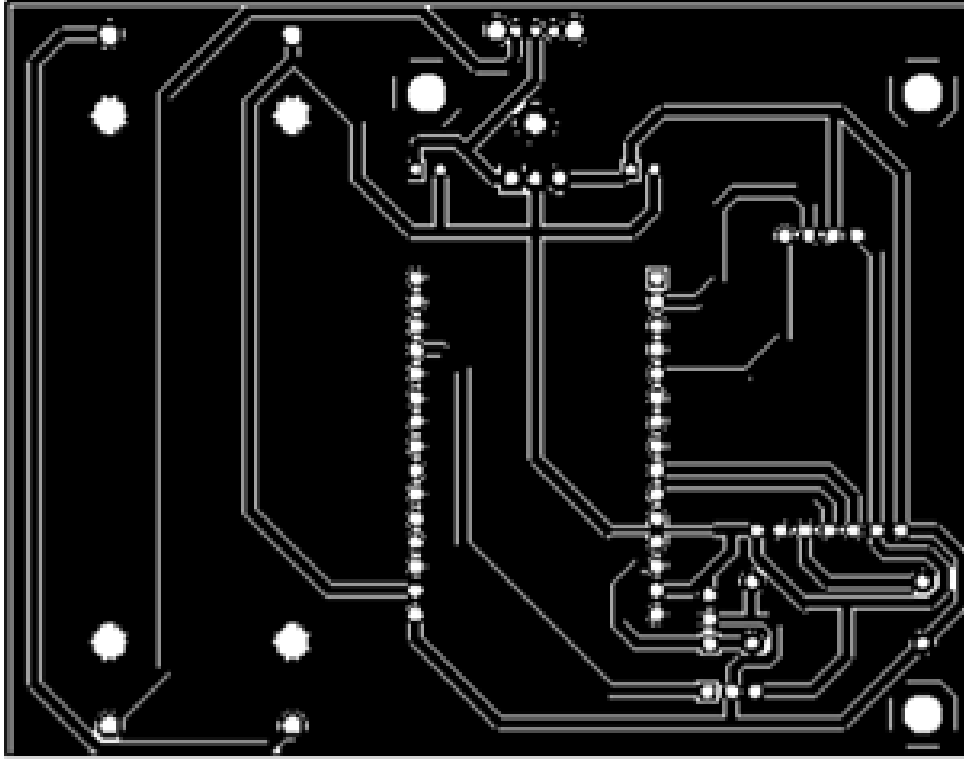


Figure 1: PCB layout design of the water quality monitoring system

Figure 2 presents a 3D rendering of the complete system, showing the physical implementation with ESP32 module, OLED display, battery holder, and all electronic components mounted on the PCB. This visualization demonstrates the compact and practical design suitable for field deployment.

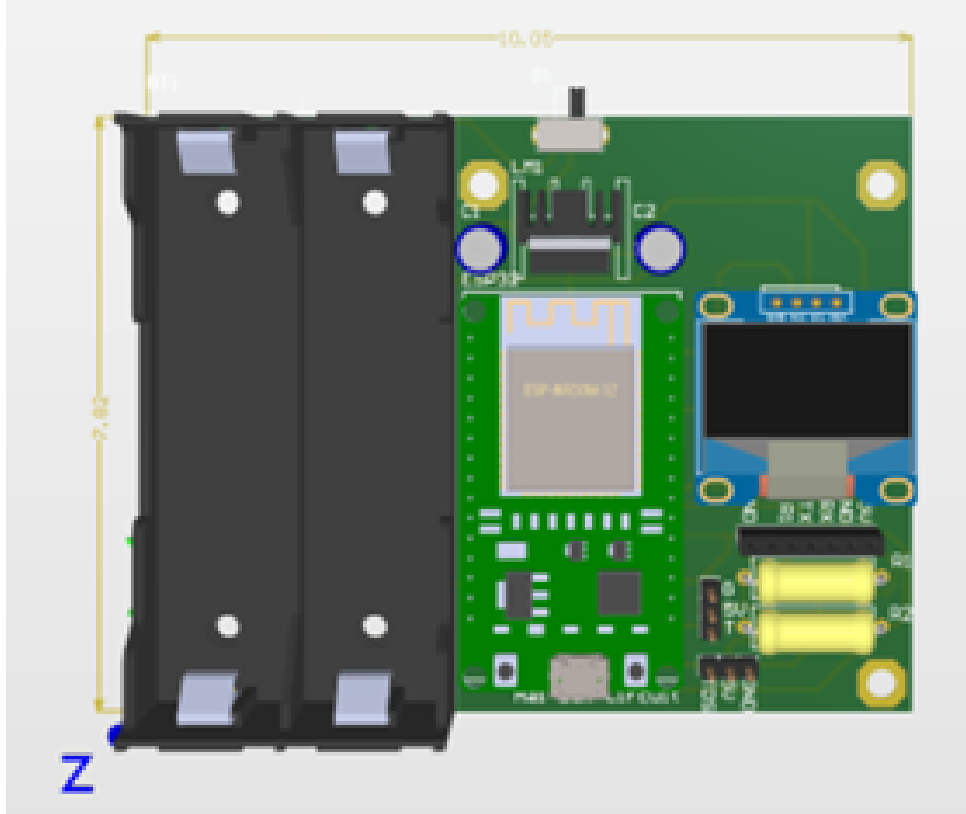


Figure 2: 3D rendering of the complete water quality monitoring system

8 Acknowledgments

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